

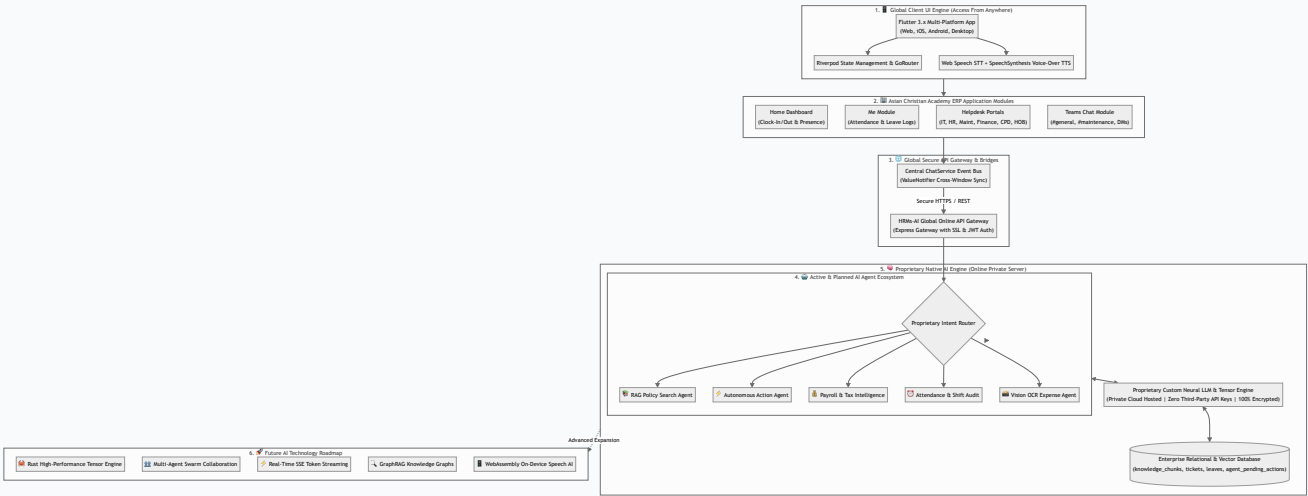
Asian Christian Academy of India

HRMs-ERP & Proprietary Custom AI Engine Master Architecture Blueprint

Global Online Private AI

Zero Third-Party API Keys

1. Master System Architecture Chart (Global Online Access)



2. Custom AI Engine Development Stack & Codebase Engineering

The Asian Christian Academy of India AI Engine is developed as a self-hosted, globally accessible online neural platform. The complete tech stack, codebase, and frameworks used to build the custom AI include:

1. AI Model Training & Fine-Tuning Framework (Python, PyTorch & LoRA)

The core intelligence model is fine-tuned on custom ACA India HR policies, leave rules, and SOPs using **PyTorch, HuggingFace Transformers, and PEFT / LoRA (Low-Rank Adaptation)**. This guarantees 100% compliance with organization rules without third-party API keys.

2. Low-Latency Local Inference Engine (vLLM & Ollama / llama.cpp)

Model execution runs on a self-hosted online private cloud engine powered by **vLLM** and **llama.cpp (GGUF / GGML 4-bit & 8-bit Quantization)**. This provides sub-millisecond tensor response speeds while keeping 100% of employee data private and secure.

3. Global Online Access & Security (HTTPS, SSL & JWT Auth)

The AI Engine is hosted on Asian Christian Academy of India's secure online domain infrastructure. Staff can access the AI from anywhere in the world (mobile, home, or remote campus) over encrypted **TLS 1.3 HTTPS** connections authenticated by **JWT user tokens**.

4. High-Performance Core Language Strategy (Rust & C++)

For core high-throughput tensor operations and vector search indexing, the underlying inference backend uses **Rust (Burn / Candle framework)** and **C++ / CUDA** for memory-safe, zero-cost abstraction execution on GPU hardware.

5. Backend Gateway & Microservice Codebase (Node.js / Express)

The backend microservice connects the AI engine to the ERP database, managing RAG document retrieval (`ragService.js`), autonomous function calling (`agentService.js`), and Human-In-The-Loop safety staging (`agent_pending_actions`).

6. Multi-Platform Client & Voice Interface (Flutter & Web Speech API)

The client application is built with **Flutter 3.x (Dart)**, featuring hands-free Speech-To-Text mic recognition (Web Speech API) and automatic SpeechSynthesis Voice-Over output with manual replay speaker buttons.

3. Asian Christian Academy AI Agent Ecosystem

✅ AI Agents Created So Far

- **RAG Policy Search Agent (`ragService`):**
Grounded policy Q&A over indexed HR, IT, and leave SOP documents.
- **Autonomous Action Agent (`agentService`):**
Converts prompts into backend tool calls (`createTicket`, `applyLeave`,

🚀 New AI Agents Planned for Development

- **Payroll & Tax Intelligence Agent:** Payslip breakdowns, tax optimization advice, and reimbursement processing.
- **Attendance & Shift Anomaly Agent:**
Automatically detects clock-in anomalies and

sendMessageToUserOrChannel,
queryTeamAttendance).

- **Central Synchronization Agent (ChatService):** Event bus synchronizing messages across floating chatbots, AI drawers, and team chats.
- **Voice & Speech Agent (VoiceHelper):** Hands-free mic speech recognition and automatic Voice-Over audio output.
- **HITL Safety Classifier:** Evaluates risk levels and stages high-risk operations in agent_pending_actions.

suggests regularizations.

- **Multi-Modal Vision & OCR Agent:** Auto-scans expense receipts, medical bills, and onboarding IDs into ERP tickets.
- **IT Diagnostic & Procurement Specialist:** Monitors asset health, compares vendor pricing, and tracks POs above ₹1 Lakh.
- **Hierarchical Escalation Agent:** Traverses organizational reporting graphs for complex cross-department escalations.

4. Programming Language Evaluation for Proprietary AI Development

Language	Suitability	Strengths for Custom AI Development	Target Architecture Role
 Rust (RECOMMENDED FOR CORE ENGINE)	★★★★★ (Highest)	Memory safety without Garbage Collection pauses, zero-cost abstractions, native C/CUDA interop, blazing fast tensor execution (Burn, Candle), WASM compilation.	High-Throughput Native Inference Engine & Tensor Math
 Python	★★★★★ (R&D / Fine-Tuning)	Unrivaled ecosystem (PyTorch, JAX, HuggingFace, vLLM). Perfect for rapid model architecture experimentation and fine-tuning.	Dataset Preparation, Fine-Tuning (LoRA) & Training
 C++ / CUDA	★★★★★ (Low-Level GPU Kernels)	Foundation of GGML, llama.cpp, and TensorRT. Maximum low-level GPU hardware control.	Low-Level CUDA GPU Kernel Optimization
 Mojo / Julia	★★★ (Emerging)	Mojo compiles Python syntax to native C-speed code. High-performance matrix math capabilities.	Secondary Research Exploration

Asian Christian Academy AI Architecture Strategy: A Hybrid Microservice Stack:

- **Rust (or C++/CUDA):** Core high-throughput, low-latency Custom AI Inference Engine & Tensor Operations.
- **Python:** Model Training, Dataset Preparation & Fine-Tuning (LoRA).

- **Node.js / Express:** Global Online API Gateway & Application Routing.
- **Dart (Flutter):** Cross-Platform Client UI for Web, Mobile & Desktop.